Making Uncertainty Visible: Methods and Tools for Hypothetical 3D Heritage Reconstructions

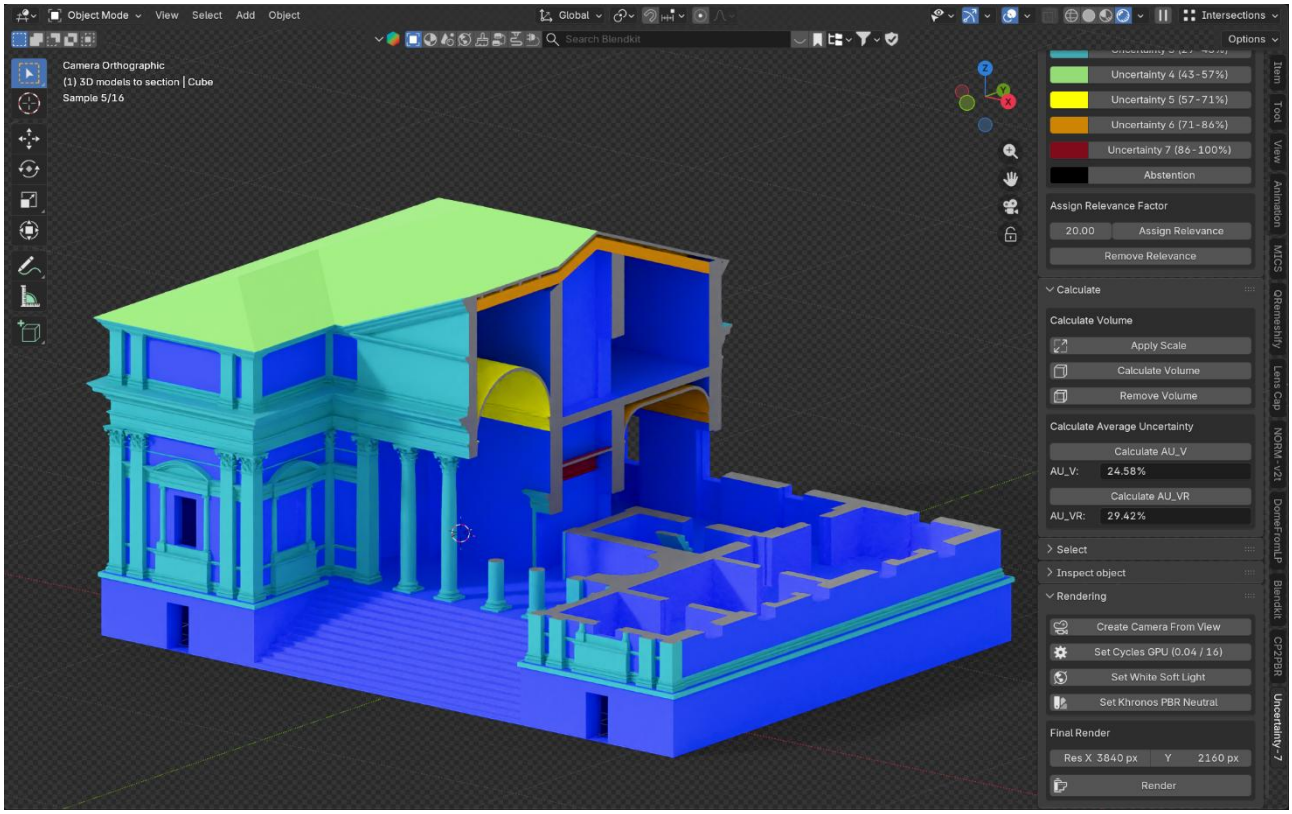


Figure: Uncertainty assessment of a hypothetical virtual reconstruction of an unbuilt Palladian Villa in Blender

The workshop offers a combined theoretical and practical exploration of the analysis and visualization of uncertainty in hypothetical 3D virtual reconstructions of architecture, with particular attention to cultural heritage and archaeological contexts.

The activity, limited to a maximum of 20 participants and requiring prior registration, will be structured as a hands-on workshop in which participants will work directly on their own laptops using free open-source tools. In particular, the open-source Blender plug-in, downloadable for free from: https://github.com/Riccardo-Foschi/Blender-Uncertainty-Calculator and developed within the Erasmus+ CoVHer project (Computer-based Visualization of Architectural Cultural Heritage), will be used.

During the workshop, participants will be shown how to download, install, and use the plug-in, which is designed to simplify and speed up the verification of geometric accuracy in models, as well as to apply the uncertainty scale developed and refined within the CoVHer project. Participants will be guided through the practical application of the proposed methodologies, learning how to integrate uncertainty into 3D models and how to communicate it effectively through three-dimensional visualization. A significant part of the workshop will be dedicated to the introduction and use of the AU_V and AU_VR formulas, including practical guidance on how to apply and interpret them in the evaluation process.

A participatory co-design session is also planned: through direct engagement with participants, the development of new uncertainty levels more specifically tailored to archaeological contexts will be initiated, collecting needs, challenges, and proposals from the scientific community.

The initiative is aimed at researchers, professionals, and students interested in digital reconstruction, heritage communication, and the development of transparent and shared methodologies for managing uncertainty in virtual models.

The goal of the workshop is to raise awareness of the importance of adopting shared best practices in the field of hypothetical heritage reconstructions, in order to ensure transparency, reproducibility, and verifiability of results, enabling the validation/falsification and reuse of digital artefacts produced by third parties. These principles align with the FAIR guidelines, which are more relevant than ever in light of the increasing accessibility of artificial intelligence tools capable of rapidly generating photorealistic results often without documentation. In this context, the risk of interpretative ambiguity and misinformation increases, as visual quality may lead users to perceive as fully reliable reconstructions that are, in fact, hypothetical. Therefore, the adoption of transparent communication strategies that clearly expose the decision-making process—inevitably based on inference and hypothesis—is more urgent and relevant than ever.

To register for the workshop, please send an email to: covher.workshop@gmail.com, indicating the workshop title and your full name.